

6. (a) The table shows the formulae of the first four members of the alcohol family.

Alcohols
CH ₃ OH
C ₂ H ₅ OH
C ₃ H ₇ OH
C ₄ H ₉ OH

Give the general formula for the alcohol family.

[1]

.....

- (b) The following word equation represents the formation of ethanol by fermentation.



- (i) Complete and balance the equation for the production of ethanol and carbon dioxide from glucose.

Include the state symbols for the reactant and the products.

[3]



- (ii) Yeast contains a substance which catalyses the breakdown of glucose.

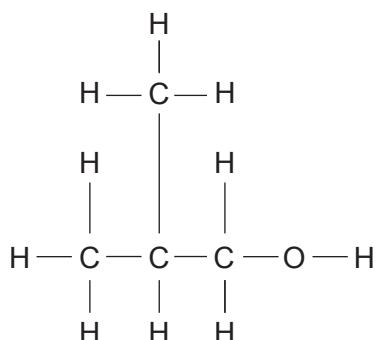
Give the **general** name of this type of catalyst.

[1]

.....



- (c) (i) C_4H_9OH has four isomers. The diagram below shows the structure of one of the isomers.



Put a **tick (✓)** in the box next to the name of this isomer.

[1]

butan-1-ol

☐

2-methylpropan-1-ol

☐

butan-2-ol

☐

2-methylpropan-2-ol

☐

- (ii) Pentan-3-ol is one of the isomers of $C_5H_{11}OH$.

Draw the structure of pentan-3-ol.

[1]

